

Nearest Neighbor Reasoning

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Apply neighbour and tie rules for Nearest neighbours and distances. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Nearest neighbours and distances. Proposed explanation: “A negative difference is a negative distance”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with $k=1$ and $k=3$. Using the same training pool and $k=3$, score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A query has nearest books A and B at distance 1, with conflicting labels; C is distance 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

6. Solve the given case

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest neighbours using differing-trait distance and $k=1$.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute distance to choose its nearest neighbour.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 22 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use neighboring liked books as evidence, while explaining which traits created closeness.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A query has nearest books A and B at distance 1, with conflicting labels; C is distance 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Calculate feature distances

For query 3 cm, calculate distances to all five training rows. Explain why A and B tie even though their labels differ.

My reasoning and evidence

13. W2 Apply neighbour and tie rules

For query 3, list the selected IDs and predictions for $k=1$ and $k=3$ using alphabetical ID order on equal distances. Explain what changes and what remains the same.

My reasoning and evidence

14. W3 Keep evaluation examples separate

Why must the three validation reference labels not enter the neighbour pool? Give the fixed $k=3$ score and baseline score on the worked validation examples.

My reasoning and evidence

15. W4 Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with $k=1$ and $k=3$. Using the same training pool and $k=3$, score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

My reasoning and evidence

16. W5 Interpret a vote and a match

Two of three neighbours vote clear. Does that prove a two-thirds chance of a correct prediction? Does an exact feature match guarantee a correct label? Explain both limits.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Nearest Neighbor Reasoning

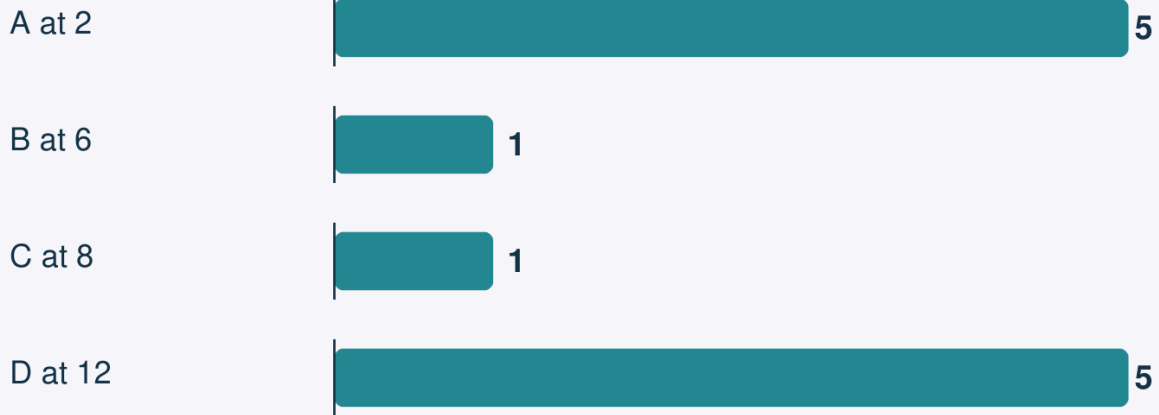
A fictional one-feature classifier predicts compact or extended book format.

Stored training rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category vote chooses compact.

Distances from query 7

absolute distance units



Bar length shows the supplied value; the marked baseline is zero.

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Why do k=1 and k=3 disagree for the original query?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Retain recommender neighbour IDs, distances, k and tie decisions so similar-book explanations can be audited.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Retain recommender neighbour IDs, distances, k and tie decisions so similar-book explanations can be audited.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional one-feature classifier predicts compact or extended book format.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B,C,A vote extended,compact,compact: predict compact.

1. $k=1$ prediction
2. Order neighbours
3. $k=3$ prediction
4. Measure distances

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: $k=3$ prediction B,C,A vote extended,compact,compact: predict compact. Changing k changes how many training neighbours vote, even though all distances remain fixed. The query's reference label must not enter the stored pool before predicting. Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-neighbour result.

Practice 2: explain the deciding evidence

Changed-case practice: Move query from 7 to 9 while keeping training fixed. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Order becomes C,B,D,A; $k=1$ predicts compact and $k=3$ predicts extended.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Order becomes C,B,D,A; $k=1$ predicts compact and $k=3$ predicts extended. Distances are $C=1, B=3, D=3, A=7$; B beats D by the ID tie rule.

Repair a missing building block

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing.. Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can 80% response be reported as 80% of the whole grade responding?. Return to this week and explain what you changed.

Review Overfitting Intuition (ai-2/16). Goal: Compare training and validation evidence for model capacity. Short practice: Does zero training error justify choosing Complex in the original record?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
